

Cholesteric Liquid Crystal Based Reconfigurable Optical Combiner for Head-Mounted Display Application

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ABSTRACT

With recent advancements in Head-Mounted Displays (HMDs), the upcoming generation of HMDs necessitates a seamless transition between Augmented Reality (AR) and Virtual Reality (VR). This paper introduces an innovative Cholesteric Liquid Crystal (CLC)-based optical combiner for HMDs, enabling dynamic switching between AR, VR, and transparent modes through temperature variation. A thermistor and external controller enable real-time temperature monitoring, which facilitates smooth transitions. We demonstrated the reconfigurability of the tunable optical combiners using 532 nm laser source. Our findings show that CLC-based optical combiners can be switched between the three modes at corresponding temperatures.

Keywords: Augmented Reality, Virtual Reality, Tunable Optics, Liquid Crystal, Wearable Electronics.

1 INTRODUCTION

Recent years have witnessed significant advancements in Head-Mounted Displays (HMDs), contributing to enhanced Human-Machine Interaction (HMI). HMDs include Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR) technologies. VR immerses users in a virtual 3D environment through headsets that block their physical surroundings. AR overlays digital content onto the real world, enhancing the user's field of view. MR combines elements of VR and AR, allowing virtual objects to interact with the real environment and enabling users to engage with both virtual and real elements simultaneously. For example, HoloLens from Microsoft employed MR technology, which combines VR and AR capabilities [4].

Despite advancements in AR, VR, and MR technology, there remains a challenge in seamlessly switching between VR, AR, and MR modes [3]. AR and MR technologies share similar hardware requirements, with optical combiners (OCs) playing a crucial role in merging virtual images with the real environment. Here, we proposed a novel liquid crystal (LC) based OC that facilitates the transition between VR, AR (MR), and transparent mode, as shown in Fig. 1. Specifically, the transparent mode allows maximum light transmission, acting like regular glasses without obstructing the view. The reconfigurable OC enables the switch from VR to AR or MR modes. This innovation aims to improve the flexibility and user experience in interacting with immersive technologies.

2 METHODS

LCs combine properties of liquids and crystalline solids, exhibiting both flowability and crystal-like characteristics. The rod-like

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structure of LC molecules allows their behavior to be controlled by factors like temperature, magnetic fields, and electric fields [1]. The behavior of LC materials can be simulated by Q-tensor theory [5]. Cholesteric Liquid Crystal (CLC) is a specific type of LC material with a helical structure, leading to selective reflection of light with certain wavelengths [2]. Our prototype used 28% w.t. S811:E7 mixture since it can reflect green light with wavelengths approximately from 500 to 540 nm at 32 °C. The selective reflection properties of CLC materials can be influenced by temperature. Fig. 1 details the relationship between CLC material and temperature.

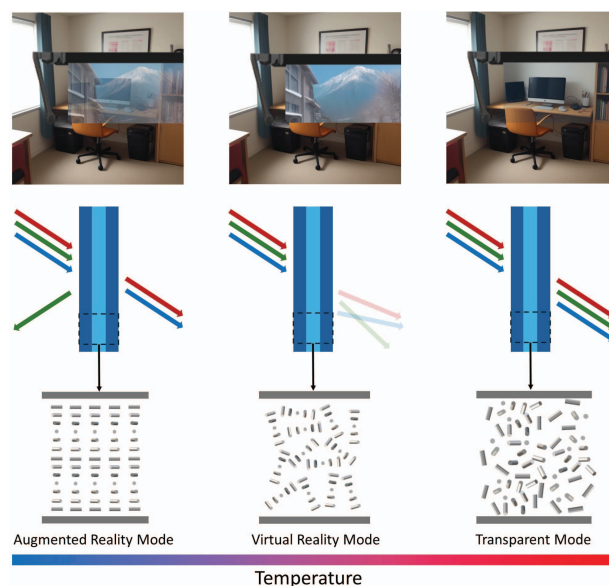


Figure 1: The schematic diagram of a reconfigurable OC switching between AR, VR, and transparent mode.

In the AR mode, the temperature is above the melting point and below the clear point of CLC material, where the CLC material exhibits selective reflection, which allows users to simultaneously see virtual images and the real environment. In the VR mode, the temperature rises to the clear point, where the liquid crystal is not fully transformed into the isotropic phase, causing a reduction in transparency and the formation of a cloudy state known as the focal conic state. In this state, the virtual image is visible and the real environment is blocked. In the transparent mode, the temperature of the OC is above the clear point, so that the CLC material is in the isotropic state and becomes fully transparent. In this paper, we demonstrated a temperature control system on a CLC cell to monitor and control the CLC temperature in real time. This system enables users to manipulate the transmittance of the reconfigurable OC and

switch different display modes on HMDs. A real-time temperature control system has been proposed to switch between different modes of CLC glass lenses. In Fig. 2a, the closed-loop system illustrates the relationship between the tunable OC and microcontroller. The CLC-based OC consists of indium tin oxide (ITO) coated glass, CLC material, and a cover glass, as shown in Fig. 2b. A DC voltage source on the ITO glass generates heat, monitored by a negative temperature coefficient (NTC) thermistor on a thin cover glass for real-time temperature feedback. The microcontroller reads the temperature and alters the voltage across the ITO glass to dynamically adjust the temperature.

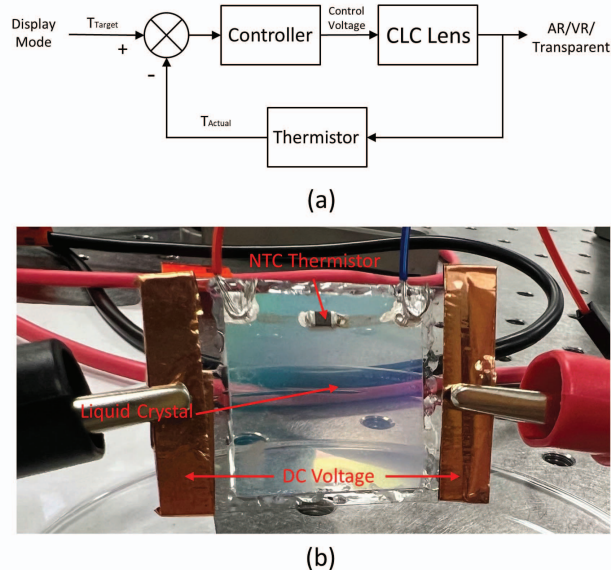


Figure 2: (a) System design of the reconfigurable OC (b) An example of a CLC-based OC.

3 RESULTS AND DISCUSSION

Firstly, the CLC-based OC was heated to 32 °C, 34 °C and 40 °C, which corresponded to AR mode, VR mode, and transparent mode, respectively. Transmittance was measured by a spectrometer, as is shown in Fig. 3a. According to the results, in AR mode, the OCs were able to block around 60% of green light from 500 to 540 nm in the environment, which could increase the contrast ratio of the virtual image. In VR mode, the OCs were opaque and the transmittance was around 10%, which could effectively block the vision of HMD users so that the user could only see the virtual image. In transparent mode, the CLC materials were in the isotropic phase and the OCs exhibited maximum transmittance.

After that, an imaging system was set up with a 532 nm laser source to simulate the light engine in HMDs and evaluate the performance of the tunable OCs. To assess the performance of the OC, measurements were taken at 32 °C, 34 °C and 40 °C. Eventually, the OC was switched between three different modes, as shown in Fig. 3b. According to the results, both the background and virtual image could be seen clearly in AR mode. As the temperature reached the clear point, the OC was in VR mode, which became opaque and only virtual images could be seen. After that, the temperature was set to 40 °C, and laser input was turned off to simulate transparent mode. In transparent mode, the OC had maximum transparency like a normal glass.

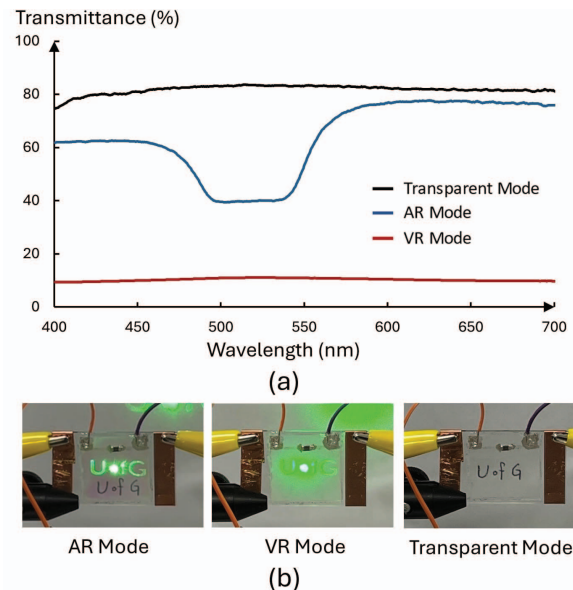


Figure 3: (a) Transmittance of the reconfigurable OC at three different modes (b) The OC switches between the three different modes at the imaging system.

4 CONCLUSIONS AND FUTURE CHALLENGES

This paper demonstrated the preliminary results of a CLC-based OC for seamless switching between AR, VR, and transparent modes in HMDs. The optic properties of CLC materials change with temperature, transitioning from the LC phase to the isotropic phase. The experimental results showcased the reconfigurability of the CLC-based OC. Notably, the OC prototype was particularly designed to reflect green light in the AR mode. Our future work is to design and fabricate OCs to reflect red and blue light in the AR mode. Subsequently, the OCs can be cascaded and block RGB light, which enables multi-color display in the system. This technology holds promise for future HMDs and head-up display applications.

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